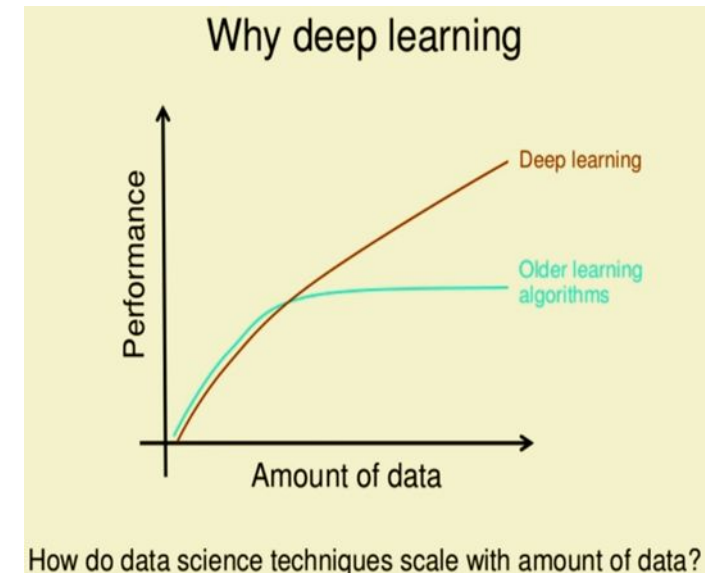


An introduction
to
Artificial Intelligence

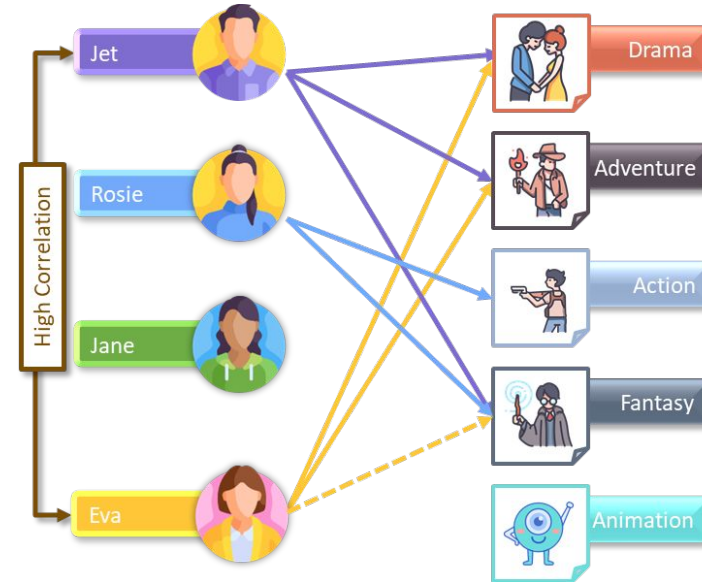
Overview

- AI, or artificial intelligence, is a branch of computer science that is designed to understand and store human intelligence, mimic human capabilities.
- AI has the ability to consume and process massive datasets and develop patterns to make predictions for the completion of future tasks.



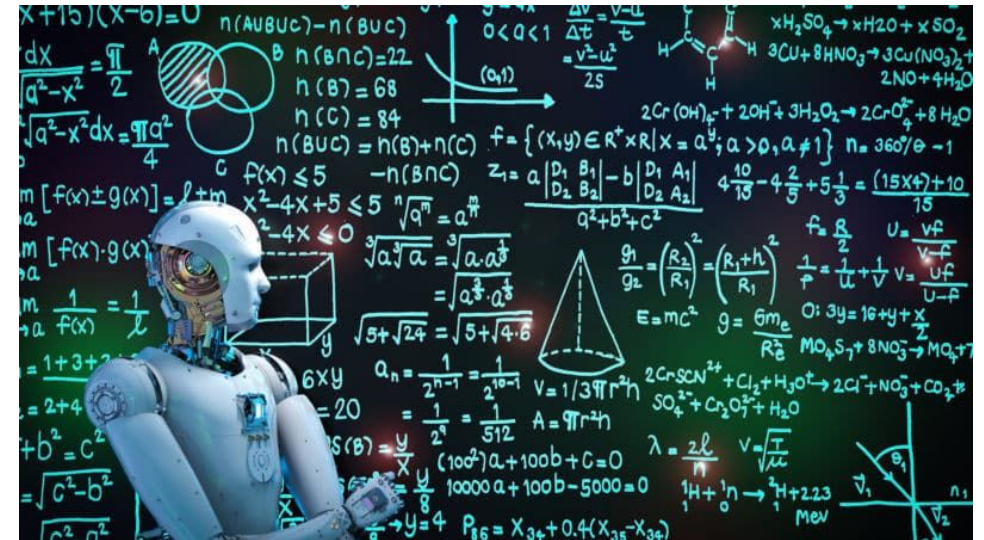
Narrow AI

- Artificial narrow intelligence – or ANI – is also known as "Weak" AI. ANI systems are capable of handling singular or limited tasks.
- Example of ANI include Apple's Siri, Netflix recommendations and the weather app where you can check the weather for the day or the week.



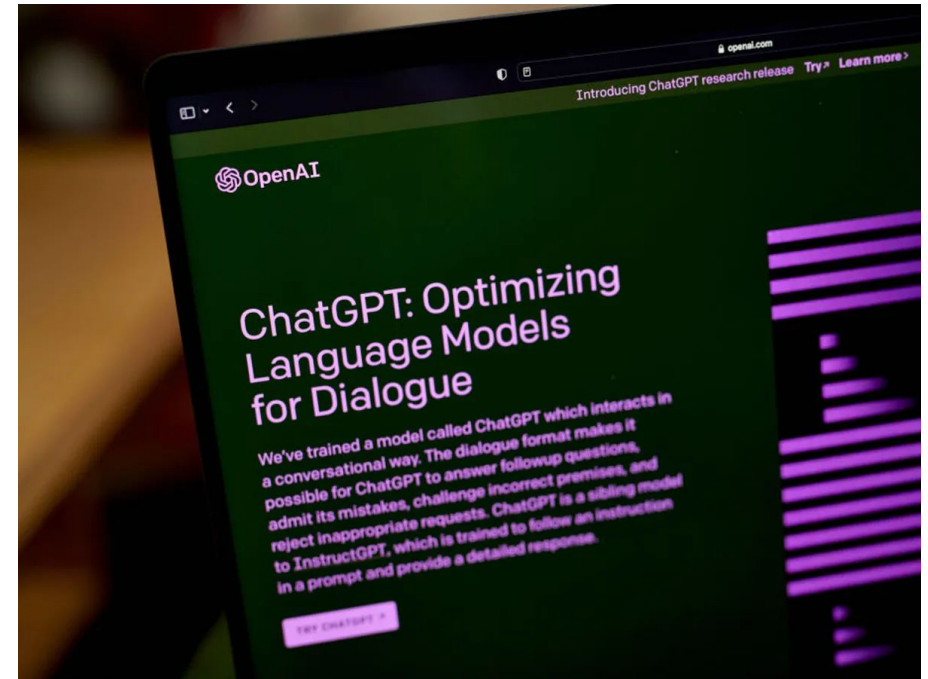
General and Super AI

- Artificial general intelligence – or AGI – is AI that can perform any intellectual task a human can.
- ASI – is the type of AI most people are fearful of. It will have the ability to surpass human intelligence in a number of ways including creativity, self-awareness, problem-solving and more.



Generative AI

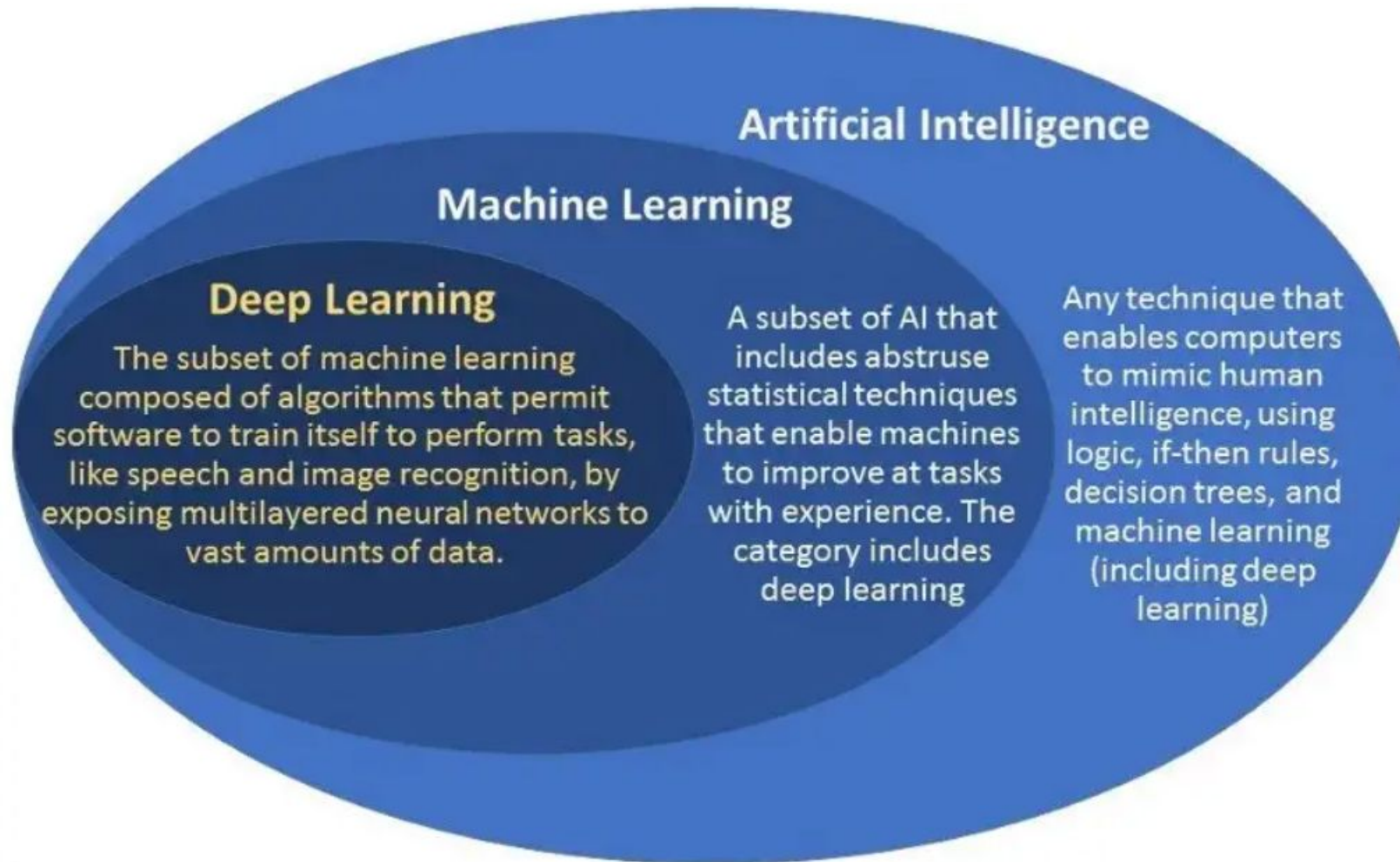
- Generative artificial intelligence is technology with the ability to produce content including text, images and more.
- Eg: writing stories or recipes, answering questions from humans, ordering products and services and more. Examples of generative AI include chatbots like ChatGPT, Bard etc.



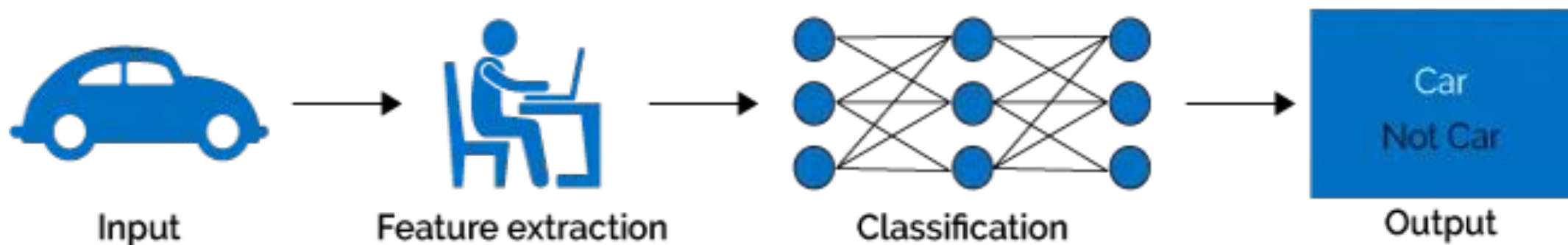
AI in our daily life

- Chatbots
- Search Engine
- Google Lens and OCR
- Social Media Feeds, Amazon, and Netflix Recommendations
- AI Image Generators
- Autonomous Vehicles
- Music and Media Streaming Services
- Video Games
- Security and Surveillance
- Facial Recognition
- Smart Speakers

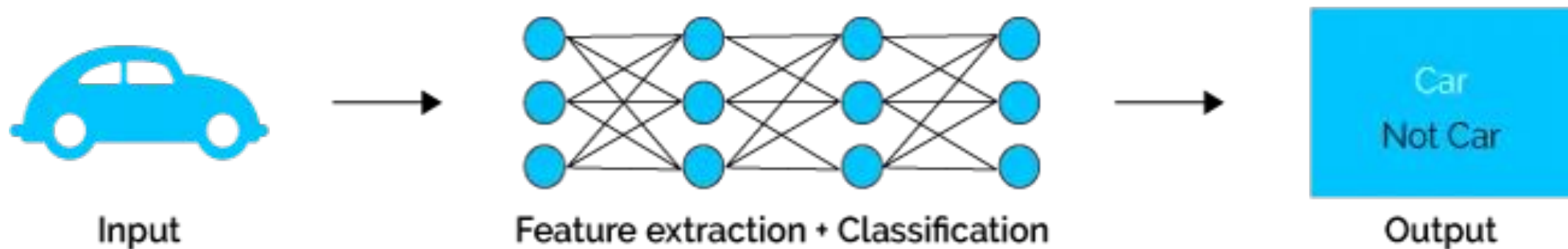
Difference between AI/ ML/ Deep Learning



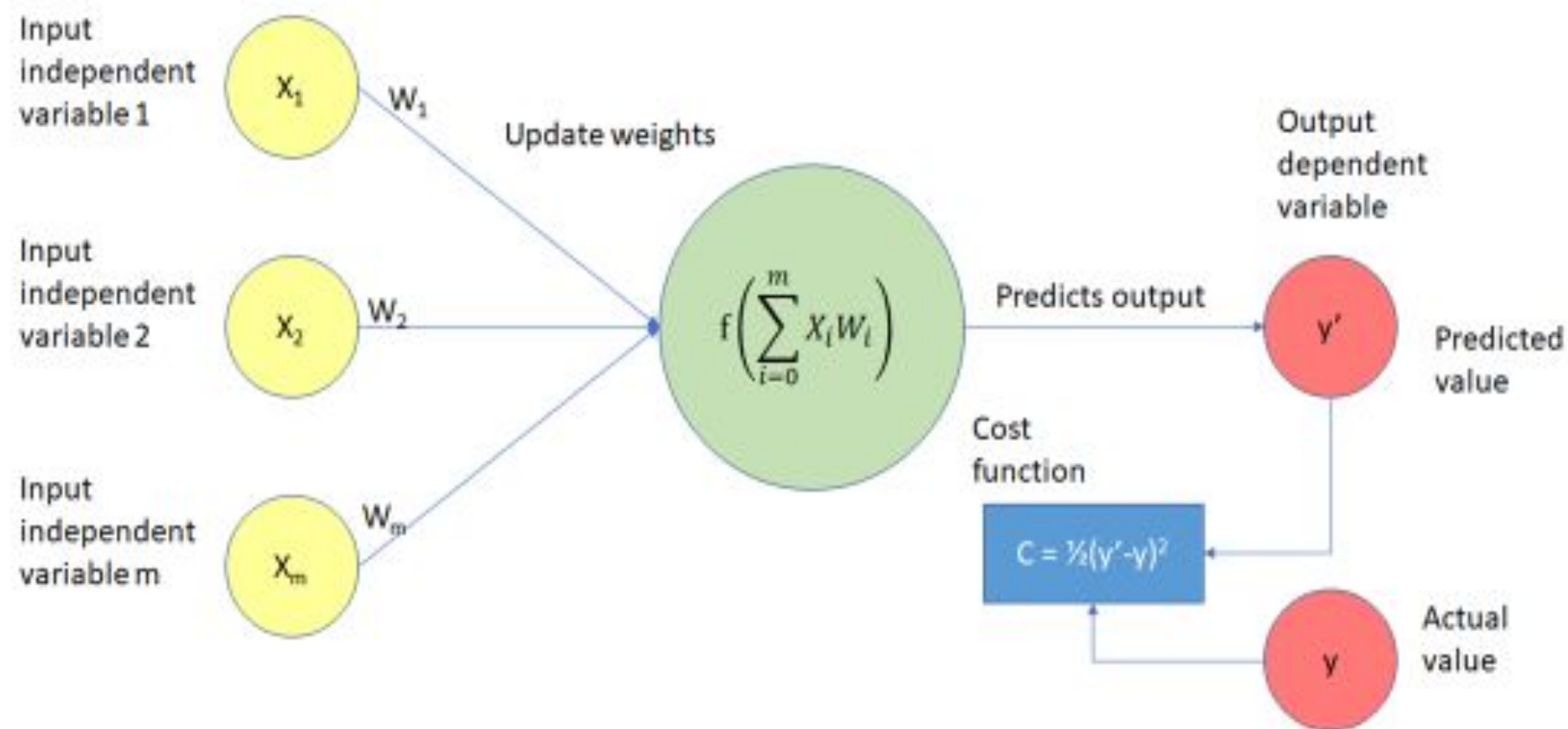
Machine Learning



Deep Learning



Working of Neural Network



Working ...

1. Calculate the error:

How far is your model output from the actual output.

2. Minimum Error:

Check whether the error is minimized or not.

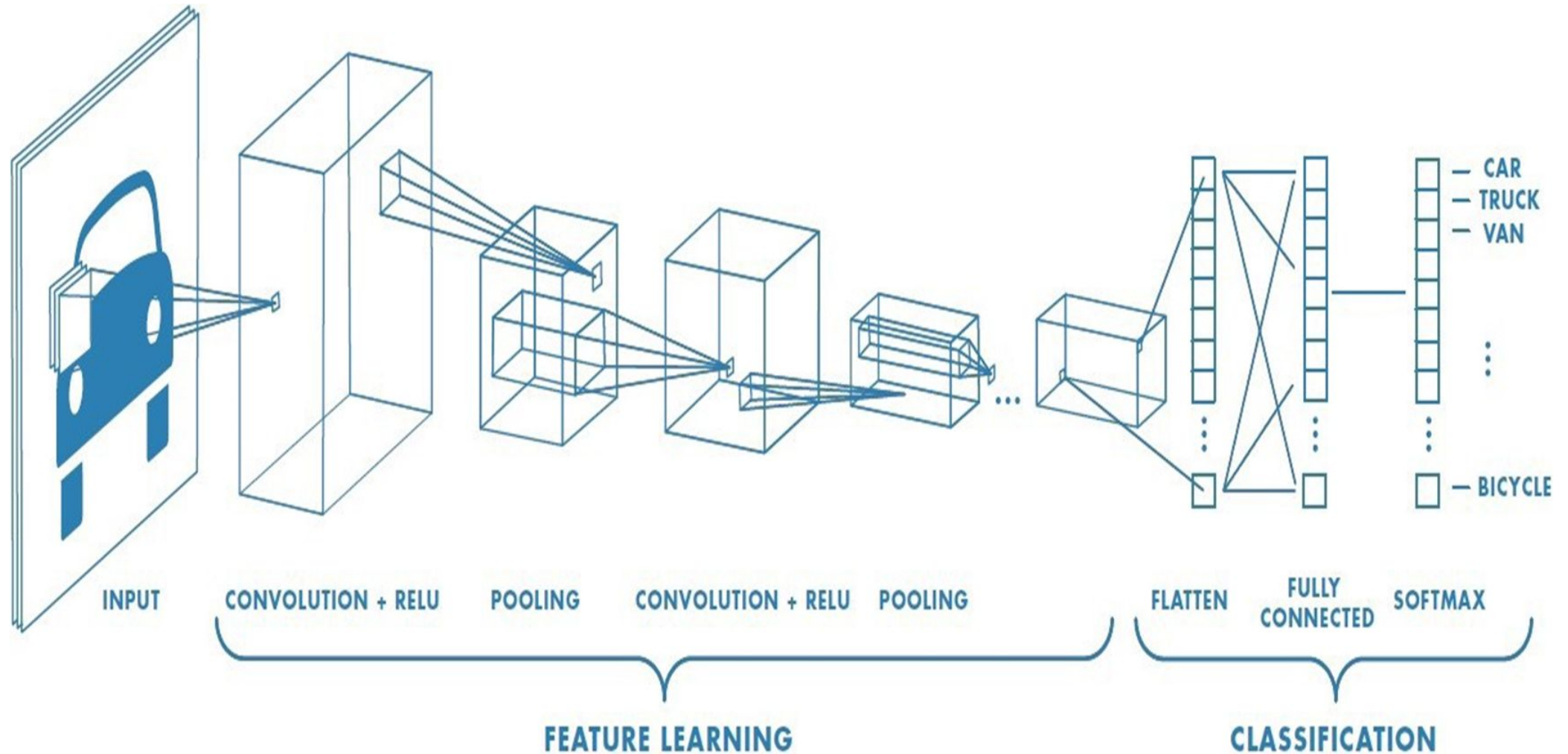
3. Update the parameters:

If the error is huge then, update the parameters (weights and biases). After that again check the error. Repeat the process until the error becomes minimum.

4. Model is ready to make a prediction:

Once the error becomes minimum, you can feed some inputs to your model and it will produce the output.

Model Architecture



How to create Deep Learning Model

1. Importing Dependencies
2. Preprocess Dataset
3. Model Creation
4. Compiling Model
5. Defining Optimizer
6. Model Training
7. Model Testing
8. Model Visualizing
9. Model Saving
10. Result Evaluation